Bridging Public Dissemination and Protected Data: A Least-Privilege SDMx Architecture on Azure Databricks

Executive Brief

1. The Question

A technology-agnostic delivery and information-governance pattern, demonstrated on Azure and Databricks.

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Statistical engineering is increasingly delivered by contractors, systems integrators and AI coding agents, yet confidentiality obligations forbid showing them the data their controls protect. This brief summarizes one worked answer for the leaders who sponsor, approve or govern that work.

The pattern fits when external specialists must implement a governed data workflow without receiving confidential production records. The institution provides an approved synthetic schema, sample lifecycle cases and an explicit access matrix, and retains production deployment, data-release and risk-acceptance decisions.

The useful test is whether the pattern reduces delivery and handover friction for a specific workload. It is not whether a synthetic demonstration proves production readiness.

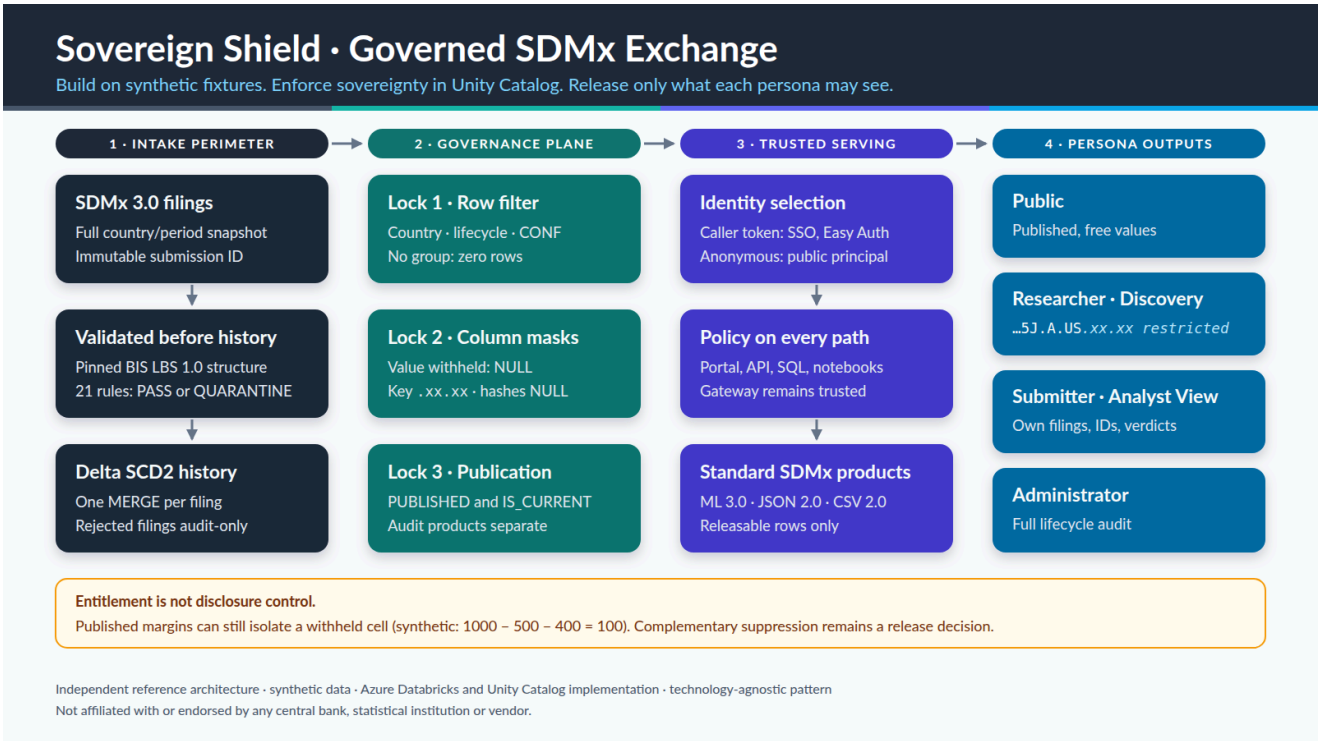
2. Who Gets What

Persona	Visible Observations	Measure Access
Public	Current published, explicitly free	Free values only
Researcher (Discovery Gateway)	Current published series across countries	Restricted values and counterparty coordinates withheld
Regional submitter	Own published/rejected filings plus public foreign observations	Own values in full; public foreign values
Administrator	All countries and lifecycle history	Full access under privileged governance

The Discovery Gateway tells a researcher that a restricted series exists, so they can approach the originating country, without revealing the value or its exact coordinates (`...US.xx.xx`). Downloads contain releasable rows only. Restricted cells would require an approved secure enclave, a documented concept. Multiple memberships combine explicitly; offboarding must remove every relevant entitlement and control-plane right.

The **Analyst View** is the regional submitter workflow. Analysts reconcile the latest filing they expect the international organization to hold against actual submission IDs, timestamps, values and validation outcomes. The latest submission and the latest accepted publication are distinct states.

3. One Architecture



Synthetic national submissions enter a protected archive. Validation makes a country/period verdict. Delta history separates accepted state from rejected arrivals. Unity Catalog evaluates query-time row and column policies. A gateway serves current SDMx feeds and distinct audit CSVs.

The international intake contract is **SDMx files only**. Synthetic bank micro-transactions are educational fixtures explaining the calculation of realistic observations; they are not an institutional intake requirement or system deliverable. Domestic granular-data collections are outside this modeled exchange.

The image describes the architecture, not proof of physical country residency or complete confidentiality protection. The demonstrator uses a shared workspace with logical jurisdiction boundaries.

4. A Revision That Matters

An accepted CA filing contains two observations: `1.111` and `2.222`. A later accepted full replacement contains only the first, revised to `3.333`.

The replacement closes both previous current records and inserts the new snapshot in one Delta transaction. The omitted observation becomes non-current, while unrelated US observations and other periods remain unchanged.

A rejected replacement does not close accepted records. Its rows retain the submission ID, submitted/received timestamps, failed observation rule and batch rejection reason. Replaying that message adds no duplicate; a genuinely new identical message remains a distinct filing.

Three decimal places are this demonstration's explicit profile, not a universal SDMx rule. Original arrivals preserve the source evidence.

5. Evidence and Limits

Demonstrated on Synthetic Data	Production Acceptance Required
Local failure injection and live UC policy-binding checks	Legacy migration and independent control acceptance
Decimal fidelity, code rejection, official JSON Schema checks	Full provisioning/content constraints and independent rule review
Local and live Delta replay and smaller-replacement checks	Distributed conflicts, recovery and production-volume measurements
Live persona SQL/API checks, public exports, provisioning and teardown	Institution-specific identity, disclosure and operational acceptance

Open challenge: published totals can still reconstruct a masked value; coordinate masking hides the counterparty, not a margin residual. Community reconstruction tests using synthetic data are invited. Restrict or remove the Researcher role if row presence makes inference trivial; public-only totals also require disclosure review. RLS and masking do not replace approved statistical disclosure control.

Detailed methods and unresolved gates are in Release Evidence. Historical screenshots are labelled historical, not fresh release results.

6. Operating Ownership

Data authority: Owns confidentiality classification, allowable use, secondary suppression, sender contracts and release approval.

Platform owner: Owns service principals, policies, deployment, backups, observability, access reviews and incident response. Resources must not depend on the contractor's personal identity.

External specialist: Builds against synthetic contracts, supplies executable tests and migration evidence, and hands over code and operational knowledge without requiring standing production access.

Independent reviewer: Approves promotion and residual risk. PR tests carry no cloud credentials; privileged plans and deployments use a protected manual gate.

Full offboarding reviews groups, sessions, tokens, Azure/vault/GitHub rights and delegated ownership. A group removal is one control, not the whole process.

The Nature of Engagement and Handover assigns repository, permission, acceptance and operational responsibilities from discovery through institution-controlled deployment.

7. Adopting the Pattern in Stages

Start with one bounded workload and widen the scope only as evidence accumulates.

Discovery: Agree the synthetic contract, access matrix, snapshot semantics, residency requirements and evaluation measures. Compare retaining the current system, extending established SDMx tooling, and an Azure-native pilot.

Bounded pilot: Implement and independently review the controls within an agreed time and spending ceiling. Default to single-node evaluation; larger compute and Photon require a separate cost decision. Measure onboarding effort, time to accepted/rejected feedback, replay recovery, query latency, and cost per accepted submission.

Assurance: Rehearse protected migration, test the live persona matrix, recovery and offboarding, validate disclosure control, and obtain legal/provenance clearance.

Reference evaluation: provisioning took approximately **42 minutes with prerequisites in place** and the deploy/test/teardown cycle cost **US$10 or less in Azure charges** (6 October 2026); the latest teardown took **32 minutes** (12 October 2026). This is an observed synthetic-workload result, not a production estimate or guaranteed price. Measurement scope.

Technology options: Terraform module/provider boundaries support AWS, GCP, Microsoft Fabric and open-source alternatives. Identity, policy enforcement, storage and lifecycle adapters must be implemented and acceptance-tested; the Azure/Databricks code is not portable unchanged.

Operational adoption: Requires accepted controls, recurring-cost estimates and an accountable operating team within the institution.

8. Questions for Discussion

Which access and revision cases would your architecture board require before permitting a pilot? Can an external contributor, human or AI agent, reproduce them without production data? Who accepts the statistical disclosure risk? Who owns the delivered runtime after the external work ends?

The contribution is a transparent integration and delivery example, complementing SDMx registries, reference infrastructure and pysdmx. It is not a claim that other organizations lack these controls or that this architecture is globally novel. Practitioners who answer these questions differently are invited to compare approaches through the repository.

Repository | Companion White Paper | Evidence and migration

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